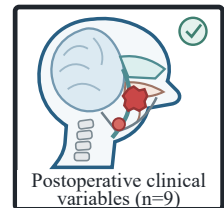


a Inputs and evidence contract

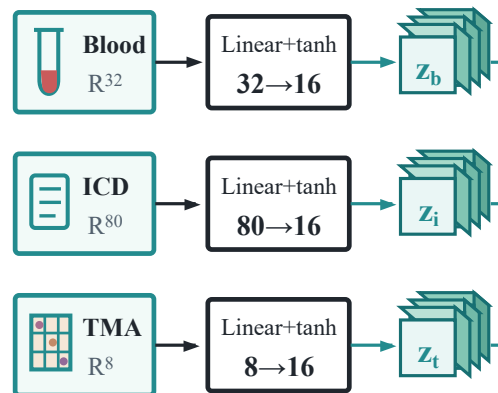
Clinical anchor



Elastic-net
Cox CAM

η_{CAM}

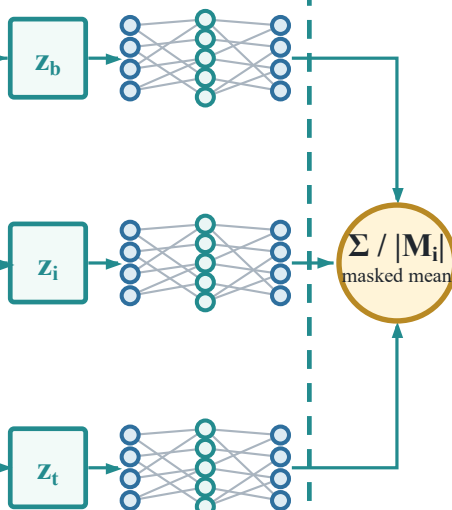
Optional modalities



Each token also receives
modality ID + status + quality

b Permutation-invariant residual

Shared ϕ : $16 \rightarrow 24 \rightarrow 16$
same weights



$\Sigma / |M_i|$
masked mean

ρ : $16 \rightarrow 16 \rightarrow 1$

$\Delta \eta_{res}$

$|M_i| = 0$

$\Delta \eta = 0$

Deep Sets · 3,225 parameters

c Nested shrinkage control

3-fold inner CV

penalty · checkpoint · λ_f
selected inside each outer fold

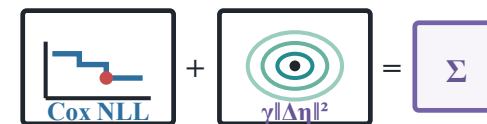
anchor
 η_{CAM}

residual
 $\Delta \eta_{res}$

Raw SCRF score

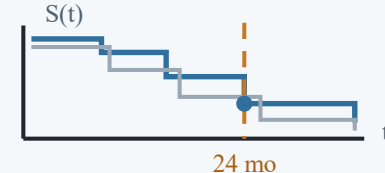
$$\eta_{SCRF} = \eta_{CAM} + \lambda_f \Delta \eta_{res}$$

Joint training objective



d Governed prediction paths

24-month survival risk



Primary raw SCRF

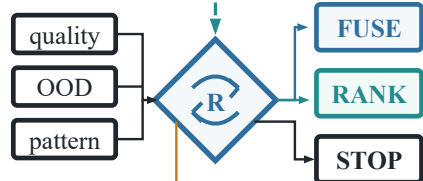
full coverage

Development-only calibration bridge

$$\alpha + 0.90 \logit(p)$$



Exploratory reliability router



CAM EXACT

— primary raw-SCRF path — optional evidence pathway — exact CAM fallback

$\oplus$ addition · $\times$ shrinkage · M_i usable modality set